Maxine Elena Calle

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Citizenship: U.S.A.

Employment and Education

Brown University — Providence, RI
Tamarkin Assistant Professor — 2027–2031
NSF Postdoctoral Research Fellow
- Postdoc mentor: Thomas Goodwillie

University of Bonn — Bonn, Germany
Hausdorff Center for Mathematics postdoctoral scholar — 2026–2027
- Postdoc mentor: Stefan Schwede

University of Pennsylvania — Philadelphia, PA
Ph.D. in Mathematics — 2026
NSF Graduate Research Fellow
- Thesis advisor: Mona Merling
- Thesis title: "Algebraic K-theory of orbispaces"

M.A. in Mathematics
- Thesis advisor: Jonathan Block
- Thesis title: "An Introduction to Symplectic Geometry for Lagrangian Floer Homology"

Reed College — Portland, OR
B.A. in Mathematics, concentration in Philosophy — 2020
Phi Beta Kappa
- Thesis advisor: Kyle Ormsby
- Thesis title: "Morse Theory and Flow Categories"

Research Interests

Algebraic topology, homotopy theory, and category theory, with a focus on higher algebraic K-theory, equivariant and stable homotopy theory, and applications to manifolds.

Papers and Publications

Research publications

- *A comparison of definitions of equivariant trees.* Joint with J.E. Bergner, D. Chan, A. Osorno, and M. Sarazola. 2025. 31 pages. Available at arXiv:2512.14956. *To appear in the Proceedings of the 2024 Mathematics Research Community on Homotopical Combinatorics.*
- *Segal K-theory factors through Waldhausen categories.* Joint with D. Chan. *Proceedings of the American Mathematical Society*: Vol. 154, No. 10 (2026), pp.4149–4166.
- *Squares K-theory and 2-Segal spaces.* Joint with M. Sarazola. *Annals of K-Theory*: Vol. 11, No. 2 (2026), pp.261–308.
- *A genuine G-spectrum for the cut-and-paste K-theory of G-manifolds.* Joint with D. Chan. 2026. 15 pages. Bulletin of the London Mathematical Society, Vol. 58, No. 4: e70352. 16 pages.
- *Equivariant Trees and Partition Complexes.* Joint with J.E. Bergner, P. Bonventre, D. Chan, and M. Sarazola. 2026. *Theory and Applications of Categories*: Vol. 45, 2026, No. 15, pp.501–536.
- *The spectrum of the Burnside Tambara functor.* Joint with D. Chan, D. Mehrle, J.D. Quigley, B. Spitz, and D. Van Niel. 2026. *International Mathematics Research Notices*: Vol. 2026, Iss. 2, rnaf388. 26 pages.

- *A linearization map for genuine equivariant algebraic K-theory.* Joint with D. Chan and A. Mejia. 2023. 50 pages. Available at arXiv:2309.08025. *To appear in Algebraic & Geometric Topology.*
- *Nested cobordisms, Cyl-objects and Temperley-Lieb algebras.* Joint with R.S. Hoekzema, L. Murray, N. Pacheco-Tallaj, C. Rovi, and S. Sridhar-Shapiro. 2025. *Topology and its Applications*: Vol. 376, 109448. 36 pages.
- *A combinatorial K-theory perspective on the edge reconstruction conjecture in graph theory.* Joint with J. J. Gould. 2025. *Homology, Homotopy, and Applications*: Vol. 27(1), pp.293–322.
- *The Spectrum of the Burnside Tambara Functor of a Cyclic Group.* Joint with S. Ginnett. 2023. *Journal of Pure and Applied Algebra*: Vol. 227, Iss. 8, 107344. 23 pages.
- *The Tambara Structure of the Trace Ideal on Cyclic Groups.* Joint with S. Ginnett, with an appendix by H. Chen and X. Chen. 2020. *Journal of Algebra*: Vol. 560, pp.114–143.
- *Sharp Sectional Curvature Bounds and a New Proof of the Spectral Theorem.* Joint with C. Dunn. 2020. *Involve, a Journal of Mathematics*: Vol. 13, No. 3, pp.445–454.
- *k-Plane Constant Curvature Conditions.* 2019. *Rose-Hulman Undergraduate Mathematics Journal*: Vol. 20, Iss. 2, Article 6. 20 pages.

Preprints

- *On the classifying space of a Morse flow category.* Joint with F. Liu. 2026. 32 pages. Available at arXiv:2603.23695. *Submitted.*
- *Equivariant algebraic K-theory for symmetric monoidal Mackey functors.* Joint with D. Chan and M. Péroux. 2023. 32 pages. Available at arXiv:2312.04705. *Submitted.*

Other publications

- *Looking into Mirror Symmetry at the 2024 JMM.* In *Feature Stories of the American Mathematical Society*, February 2024.
- *Bringing Math into The Conversation: My Summer as a Mass Media Fellow.* In *Notices of the American Mathematical Society*, February 2024, pp.274–275.
- *A brief illustrated guide to 'scissors congruence' – an ancient geometric idea that's still fueling cutting-edge mathematical research.* Co-authored with M. Merling. 2023. *The Conversation U.S.*
- *Proving Fermat's last theorem: 2 mathematicians explain how building bridges within the discipline helped solve a centuries-old mystery.* Co-authored with D. Bressoud. 2023. *The Conversation U.S.*

Fellowships, Grants, and Awards

Academic and research

- Oberwolfach Research Fellows program on "Derived orbifold bordism" 2027
  Oberwolfach Research Institute for Mathematics
- Carlitz–Zippin Prize for an exceptional Ph.D. thesis 2026
  University of Pennsylvania, Mathematics Department
- Mathematical Sciences Postdoctoral Research Fellowship 2026
  National Science Foundation
  Award deferred (1 year) for a postdoctoral appointment at the University of Bonn
- AIM SQuaREs grant for "Algebras over genuine equivariant operads" 2025
  American Institute of Mathematics
- AIM SQuaREs grant for "An algebraic approach to Thom spectra" 2025
  American Institute of Mathematics
- AIM SQuaREs grant for "Two-dimensional nested cobordisms and TQFTs with substructure" 2024
  American Institute of Mathematics
- Herbert S. Wilf Memorial Award for excellence in research and communication 2023-2024
  University of Pennsylvania, Mathematics Department
- Dean's Scholar 2023
  University of Pennsylvania, School of Arts & Sciences
- Graduate Research Fellowship 2020–2025
  National Science Foundation

- Calabi Scholar 2020–2025
  University of Pennsylvania, Mathematics Department
- Benjamin Franklin & Fontaine Fellowship 2020–2025
  University of Pennsylvania
- Phi Beta Kappa 2020
  Reed College
- Commendation for Academic Achievement 2016-2020
  Reed College

Teaching and pedagogy

- Moez Alimohamed Graduate Student Teaching Award for excellence in teaching 2026
  University of Pennsylvania, Mathematics Department
- Prison Teaching Initiative Graduate Fellow 2024-2025
  Prison Teaching Initiative at Princeton University
- CETLI Teaching Certificate 2024
  Center for Teaching and Learning at the University of Pennsylvania
- Master TA 2023-2026
  University of Pennsylvania, Mathematics Department
- Inclusive & Equitable Teaching Fellow 2023-2024
  Center for Teaching and Learning at the University of Pennsylvania
- Good Teaching Award, Math 3700 Fall 2022
  University of Pennsylvania, Mathematics Department

Service, communication, and outreach

- Postdoctoral/Advanced Graduate Ambassadorship 2023-2024
  Lisa Simonyi and the Women and Mathematics Program at the Institute for Advanced Study
- Mass Media Fellow 2023
  American Association for the Advancement of Science
  Sponsored by the American Mathematical Society and hosted by The Conversation U.S.
- Penn Graduate Community-Engagement Research Fellow Summers 2021 and 2022
  The Netter Center at the University of Pennsylvania
- Elevating Mathematics competition, honorable mention (joint with L. Williams) 2020
  National Academies of Sciences, Engineering, and Medicine
- MAA MathFest Outstanding Presentation Award 2019
  Joint Mathematics Meetings

Teaching Experience

The University of Pennsylvania Philadelphia, PA

Instructor
Math 3120: Linear algebra Summer 2025

Teaching Assistant
Math 3700: Algebra Spring 2026
Math 2600: Honors Calculus II Spring 2023
Math 3700: Algebra Fall 2022

Grader
Math 6180: Algebraic Topology I Fall 2025

Workshop Facilitator
Trauma-Informed Pedagogy Spring 2025
Inclusive & Equitable Teaching in Mathematics (mini-course) Fall 2023
Mathematics Department TA Training Falls 2023–2025
School of Arts & Sciences Graduate TA Training: Inclusive & Equitable Teaching at Penn Fall 2023

Reed College Portland, OR

Individual Tutor and Drop-in Tutor at the Math Help Center Spring 2017–Spring 2020
Math 111: Calculus, Math 112: Introduction to Analysis, Math 113: Discrete Structures,
Math 201: Linear Algebra, Math 202: Vector Calculus, Math 311: Complex Analysis,
Math 321: Real Analysis, and Math 332: Abstract Algebra

Course Assistant Fall 2018–Spring 2020
Math 202: Vector Calculus

Prison teaching

Co-instructor at State Correctional Institution Chester Chester, PA
Mathematics in the Courtroom Spring 2026

Co-instructor at South Woods State Prison Bridgeton, NJ
Math-015: Beginning Mathematics Fall 2024
Math-015: Beginning Mathematics Spring 2022

Co-instructor at Garden State Youth Correctional Facility Crosswicks, NJ
Math-015: Beginning Mathematics Fall 2021

Mathematics Tutor at South Woods State Prison Bridgeton, NJ
Math-010: Number Systems, Math-015: Beginning Mathematics, 2023–2025
Math-020: Elementary Algebra

Course Development team leader
Math-015: Beginning Mathematics Summer 2024

Other Experience and Service

Seminar and conference organization

- **Talbot Workshop** 2023–2026
  - Talbot 2026: Topological holography
    co-organizer with A. Karapetyan, C. Li, and S. Xue
  - Talbot 2025: Homological stability,
    co-organizer with A. Karapetyan and E. Sukarto
  - Talbot 2024: Topological cyclic homology of ring spectra,
    co-organizer with A. Holeman, A. Karapetyan, L. Keenan, and E. Sukarto
  - Talbot 2023: Computations in stable motivic homotopy theory,
    co-organizer with A. Holeman, L. Keenan, E. Sukarto, and L. Yang
- **Atlantic Meeting on Topology, Representation theory, and K-theory (AMTRaK),** 2024–2025
  co-organizer with A. Chedalavada and B. Spitz
  - Homological stability, at the University of Pennsylvania March-April 2025
  - Hermitian K-theory, at the University of Virginia February 2025
  - Dualizable categories and continuous K-theory, at Johns Hopkins University November 2024
  - Equivariant and parametrized homotopy theory, at the University of Pennsylvania September 2024
- **Gender Minorities in Topology and Related Areas Konference (GeMTRAK)** April 2024
  at the University of Pennsylvania, *co-organizer with M. Ong and H. Stufflebeam*
- **Graduate Geometry/Topology Seminar** at the University of Pennsylvania, 2022–2023
  co-organizer with A. Mejia
- **Graduate Pizza Seminar** at the University of Pennsylvania, *co-organizer with K. Canzius* 2021–2022
- **UP GRADe** workshop at the University of Pennsylvania, *co-organizer with* October 2021
  A. Gibney, J. Hartmann, M. Merling, M. Ong, and E. Urheim

Community building and mentorship

- **Directed Reading Program** at the University of Pennsylvania, *co-organizer with*
  L. Ferreira Guilhoto (Fall 2023–Spring 2025) and mentor for the following projects:
  - Shotaro Hiranuma, *Continuing in category theory* Spring 2025
    Category theory and logic Fall 2024
  - Talia Becker Calazans, *Combinatorial topology* Fall 2023
  - Elena Isasi Theus, *Algebraic topology and the mathematical basis of topological data analysis* Spring 2023
    Fundamentals of computational applied topology Fall 2022
  - Cianán Conefrey-Shinozaki, *2D topological quantum field theories* Spring 2022

– Michael Zeng, *Simplicial homotopy theory* Fall 2021

- **Prison Education Program** at Eastern University, *volunteer* Spring 2024
- **Prison Teaching Initiative** at Princeton University, *volunteer* 2021–2025
- **GeMs in Math** group at the University of Pennsylvania, *co-coordinator* 2021–2025
- **STEMGeMs** at Reed College, *co-leader and department liaison* 2019–2020
- **Peer Mentor Program** at Reed College, *mentor* 2017-2018

Science communication, outreach, and other service

- **Penn Grad Talks** at Penn Arts & Sciences Spring 2026
  Natural Sciences finalist
- **Math Circles** at West Philadelphia High School Spring 2024
  Volunteer
- **The Conversation U.S.** 2023
  Freelancer (Fall 2023) and AMS-AAAS 2023 Mass Media Fellow (Summer 2023)
- **Math Circles in West Philadelphia** project at the University of Pennsylvania Summer 2022
  Graduate fellow, joint project with M. Ong, mentored by M. Merling and Y. Ou Philadelphia, PA
- **Mathematics Enrichment and Academically-Based Community Service** project at the University of Pennsylvania Summer 2021
  Graduate fellow mentored by M. Merling Philadelphia, PA
- **Referee and reviewer**
  Advances in Mathematics, Proceedings of the AMS, PUMP Journal of Undergraduate Research, MathSciNet, zbMATH Open

Selected Research Workshops

- **CMO Workshop** at Casa Matemática Oaxaca September 2025
  Invited participant in "Equivariant Topological Quantum Field Theory" Oaxaca, MX
- **AIM Workshop** at American Institute of Mathematics July 2024
  Invited participant in "Scissors congruences, algebraic K-theory and Steinberg modules" Caltech, CA
- **BIRS Workshop** at the Banff International Research Station January 2024
  Invited participant in "Higher Segal Spaces and their Applications" Banff (Canada)
- **BIRS Workshop** at the Banff International Research Station November 2023
  Invited participant in "Spaces of Manifolds: Algebraic and Geometric Approaches" Banff (Canada)
- **Women in Topology IV** at the Hausdorff Institute August 2023
  Team member for research project on cobordism categories Bonn (Germany)
- **RTG Collaborative Workshop in Homotopy Theory** August 2022
  Team member for research project on equivariant homotopy theory Lexington, VA
- **Collaborative Mathematics Research Group** at Reed College Summer 2019
  Participant in undergraduate research program on algebraic topology Portland, OR
- **NSF REU** program at California State University, San Bernardino Summer 2018
  Participant in undergraduate research program on differential geometry San Bernardino, CA

Selected Talks and Presentations

Research conference talks

Conference of the French Algebraic Topology Network Grenoble, France
Cut-and-paste K-theory of manifolds and SK-automorphisms October 2026

AMS 2025 Fall Eastern Sectional Meeting, Special Session: Homotopy theory St. Louis, MO
Segal K-theory factors through Waldhausen categories October 2025

CMO Workshop: Equivariant Topological Quantum Field Theory Oaxaca, Mexico
Cut-and-paste K-theory of manifolds and cobordism September 2025

Mathematical Congress of the Americas, Special Session: Interactions of Miami, FL
equivariant bordism and low dimensional topology July 2025
Towards algebraic K-theory of orbifolds

Joint Mathematics Meetings, Special Session: Homotopical combinatorics Seattle, WA

The Edge Reconstruction Conjecture and Combinatorial K-theory — January 2025

Binghamton University Graduate Combinatorics, Algebra, and Topology Conference (BUGCAT) — Binghamton, NY
The Edge Reconstruction Conjecture and Combinatorial K-theory — October 2024

AMS 2024 Fall Eastern Sectional Meeting, Special Session: Homotopy theory and K-theory — Albany, NY
Squares K-theory and 2-Segal spaces — October 2024

Graduate Student Conference in Algebra, Geometry, and Topology — Philadelphia, PA
Equivariant Trees and Partition Complexes — June 2024

Joint Mathematics Meetings, Special Session: Equivariant techniques in stable homotopy theory — San Francisco, CA
A linearization map for equivariant A-theory — January 2024

Joint Mathematics Meetings, Special Session: Topics in equivariant algebra — San Francisco, CA
Ideals in the Burnside Tambara functor on a cyclic group — January 2024

Binghamton University Graduate Combinatorics, Algebra, and Topology Conference (BUGCAT) — Binghamton, NY
Equivariant Partition Complexes and Trees — November 2022

Nebraska Conference for Undergraduate Women in Mathematics — Lincoln, NE
The Tambara Structure of the Trace Ideal — January 2020

Undergraduate Mathematics Symposium — Chicago, IL
Putting the 'k' in Curvature: k-Plane Constant Curvature Conditions — November 2018

Northwest Undergraduate Mathematics Symposium — Salem, OR
Putting the 'k' in Curvature: k-Plane Constant Curvature Conditions — November 2018

Invited seminar talks

HSM Colloquium — Bonn, Germany
How to cut things up (and put them back together again) using algebraic topology — November 2026

University of Bonn Topology Seminar — Bonn, Germany
On the scissors congruence K-theory of cut-and-paste manifolds — October 2026

MPIM Topology Seminar — Bonn, Germany
Flavors of Higher Algebraic K-theory (short talk) — October 2026

eCHT Research Seminar — Online
Prime ideals in the Burnside Tambara functor — September 2026

MPIM Abstract Homotopy Theory Seminar — Bonn, Germany
Group-completion K-theory via Waldhausen categories — September 2026

Texas State Topology Seminar — San Marcos, TX
Classifying spaces and Morse flow categories — May 2026

University of New Mexico Geometry & Topology Seminar — Las Cruces, NM
Towards algebraic K-theory of orbifolds — March 2026

University of Pennsylvania Geometry & Topology Seminar — Philadelphia, PA
Nested cobordisms and TQFTs — February 2026

University of Copenhagen Algebra and Topology Seminar — Online
Towards algebraic K-theory of orbifolds — February 2026

Topological Quantum Field Theory Club Seminar — Online
Nested cobordisms and TQFTs — January 2026

University of Indiana Bloomington Topology Seminar — Bloomington, IN
Cut-and-paste K-theory of manifolds and SK-automorphisms — December 2025

Universität Bonn Topology Seminar — Bonn, Germany
Towards algebraic K-theory of orbifolds — November 2025

Brown University Topology Seminar — Providence, RI
Cut-and-paste K-theory of manifolds and SK-automorphisms — November 2025

PIMS Collaborative Research Group "Diagram Categories in Homotopy Theory" Seminar — Online
Cut-and-paste K-theory of manifolds and SK-automorphisms — October 2025

University of Pennsylvania Geometry & Topology Seminar — Philadelphia, PA
Towards algebraic K-theory of orbifolds — February 2025

University of Chicago Topology Seminar — Chicago, IL
Towards A-theory of orbifolds — January 2025

Northwestern University Topology Seminar — Evanston, IL
Towards A-theory of orbifolds — January 2025

University of Virginia Topology Seminar — Charlottesville, VA
Towards A-theory of orbifolds — November 2024

Johns Hopkins University Topology Seminar — Baltimore, MD
Towards A-theory of orbifolds — October 2024

Texas State Topology Seminar — San Marcos, TX
A linearization map for equivariant A-theory — March 2024

University of Indiana Bloomington Topology Seminar — Bloomington, IN
Nested Cobordisms and TQFTs — March 2024

University of Minnesota Topology Seminar — January 2024
All connective G-spectra come from K-theory — Minnesota, MN

Loyola University TACO Seminar — Chicago, IL
Equivariant Partition Complexes and Trees — November 2023

Michigan State University RTG Seminar — East Lansing, MI
Nested Cobordisms and TQFTs — November 2023

SUNY Binghamton Geometry and Topology Seminar — Binghamton, NY
Nested Cobordisms and TQFTs — September 2023

UCLA Algebraic Topology Seminar — Los Angeles, CA
Equivariant Partition Complexes and Trees — May 2023

UCSD Topology Seminar — San Diego, CA
Equivariant Partition Complexes and Trees — May 2023

Purdue University Topology Seminar — Lafayette, IN
Equivariant Partition Complexes and Trees — April 2023

University of Rochester Topology Seminar — Rochester, NY
Equivariant Partition Complexes and Trees — April 2023

University of Virginia Topology Seminar — Charlottesville, VA
Equivariant Partition Complexes and Trees — February 2023

Invited expository talks

Penn Arts & Sciences Penn Grad Talks — Philadelphia, PA
What is math good for, anyways? (short talk) — March 2026

Moduli spaces of pseudo-holomorphic curves workshop — Otego, NY
Orbispaces as stacks: geometry and examples — May 2025

SLMath Hot Topics Workshop: Life after the Telescope Conjecture — Berkeley, CA
Cyclotomic spectra — December 2024

Talbot 2024: Topological cyclic homology of ring spectra — Nacogdoches, TX
THH as a cyclotomic spectrum — August 2024

Scissors congruences, algebraic K-theory and Steinberg modules AIM Workshop — Pasadena, CA
The scissors congruence K-theory of polytopes is a Thom spectrum — July 2024

Spaces of Manifolds: Algebraic and Geometric Approaches BIRS seminar — Online
The stable parametrized h-cobordism theorem: the K-theory part — October 2023

Scissors Congruence, Algebraic K-Theory, and Trace Methods summer school — Bloomington, IN
Cut-and-paste K-theory of manifolds and square K-theory — June 2023

Talbot 2022: Scissors congruence and algebraic K-theory — New Lebanon, NY
Square K-theory and Manifold Invariants — June 2022

Talks for undergraduates

UPGrade Workshop — Philadelphia, PA
How to cut up shapes using algebraic topology (short talk) — May 2026

Reed College Mathematics Colloquium — Portland, OR
The Edge Reconstruction Conjecture and Combinatorial K-theory — October 2024

Providence College Mathematics Colloquium — Providence, RI
The Edge Reconstruction Conjecture and Combinatorial K-theory — September 2024

Temple University Math Club — Philadelphia, PA
Scissors Congruence: how to cut up shapes and get away with it — March 2024

Penn Undergraduate Mathematics Society (PUMS) speaker series — Philadelphia, PA
Social choice problems and algebraic topology (short talk) — April 2024
Configurations in the plane (short talk) — November 2023
Scissors Congruence: how to cut up shapes and get away with it — December 2022

Penn Graduate Student Open House lightning talks — Philadelphia, PA
A quick introduction to cut-and-paste manifolds — February 2026
A quick introduction to cut-and-paste manifolds — February 2025
Social choice problems and algebraic topology — February 2024